Materials & Solid State Physics Notes

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May 2026 -

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1 Quantum Mechanics

1.1 Pauli Exclusion Principle

In quantum mechanics, the information about a particle is not local to a single point in any measurable space. This spread in the information of a quantum-mechanical particle tells us that if we have two identical particles such as electrons, we will unable to assign a consistent label to each particle due to their wavefunction overlap. In classical physics, if we assign labels to two objects and track their trajectories over time, we will always be able to identify the path taken by the separate objects, which allows them to be labeled, or distinguishable. The *indistinguishability* of quantum particles is thus a significant deviation from classical physics.

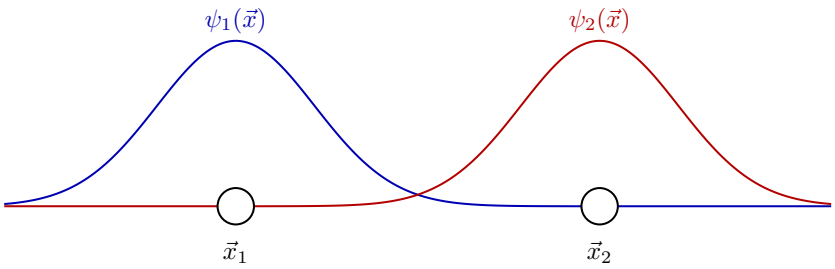


Figure 1: Visualization of how wavefunction overlap gives rise to the indistinguishability of two quantum-mechanical particles.

If we therefore have a two-particle wavefunction, where both particles are given a specific position label,

$$\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2), \tag{1}$$

we can arbitrarily swap the position labels

$$(\boldsymbol{x}_1, \boldsymbol{x}_2) \rightarrow (\boldsymbol{x}_2, \boldsymbol{x}_1) \tag{2}$$

without any *measurable* change to the wavefunction:

$$\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) \rightarrow e^{i\theta}\Psi(\boldsymbol{x}_2, \boldsymbol{x}_1). \tag{3}$$

The phase factor can be added since it doesn't add any measurable change to the wavefunction (global phases don't change probabilities). To determine

how the phase shift changes the nature of the wavefunction, we apply this operation twice:

$$\begin{aligned}\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) \to e^{i\theta}\Psi(\boldsymbol{x}_2, \boldsymbol{x}_1) \to e^{2i\theta}\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) \\ \implies e^{2i\theta} = 1 \\ \implies \theta = n\pi \\ \implies e^{i\theta} = \pm 1\end{aligned} \tag{4}$$

This tells us that under label exchange,

$$\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = \pm\Psi(\boldsymbol{x}_2, \boldsymbol{x}_1). \tag{5}$$

If we construct a general two-particle state, we must therefore consider the two separate label choices:

$$\boxed{\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = \frac{1}{\sqrt{2}}\Big(\psi_1(\boldsymbol{x}_1)\psi_2(\boldsymbol{x}_2) \pm \psi_1(\boldsymbol{x}_2)\psi_2(\boldsymbol{x}_1)\Big)} \tag{6}$$

From quantum field theory, spin-$\frac{1}{2}$ particles, called fermions, take the $-$ sign and spin-1 particles, called bosons take the $+$ in this exchange operation.

For bosons,

$$\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = \frac{1}{\sqrt{2}}\Big(\psi_1(\boldsymbol{x}_1)\psi_2(\boldsymbol{x}_2) + \psi_1(\boldsymbol{x}_2)\psi_2(\boldsymbol{x}_1)\Big), \tag{7}$$

and if we set $\psi_1 = \psi_2 \equiv \psi$ we get,

$$\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = \psi(\boldsymbol{x}_1)\psi(\boldsymbol{x}_2). \tag{8}$$

This tells us that *identical bosons can occupy the same state* ψ.

For fermions,

$$\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = \frac{1}{\sqrt{2}}\Big(\psi_1(\boldsymbol{x}_1)\psi_2(\boldsymbol{x}_2) - \psi_1(\boldsymbol{x}_2)\psi_2(\boldsymbol{x}_1)\Big). \tag{9}$$

We cannot use the trick as before by setting $\psi_1 = \psi_2 \equiv \psi$, for if we do, we get

$$\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2) = \frac{1}{\sqrt{2}}\Big(\psi(\boldsymbol{x}_1)\psi(\boldsymbol{x}_2) - \psi(\boldsymbol{x}_2)\psi(\boldsymbol{x}_1)\Big) = 0. \tag{10}$$

This tells us that, unlike bosons, *fermions cannot occupy the same state* ψ. This is the **Pauli exclusion principle**.

The Pauli exclusion principle states that two or more spin-$\frac{1}{2}$, called fermions, cannot occupy the same state.

The fermionic case of Eq. (6) can be written as a two-particle *Slater determinant*, which is a neat trick for writing the full wavefunction of a multi-fermionic system. In general,

$$\Psi(\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_1(\boldsymbol{x}_1) & \psi_2(\boldsymbol{x}_1) & \ldots & \psi_N(\boldsymbol{x}_1) \\ \psi_1(\boldsymbol{x}_2) & \psi_2(\boldsymbol{x}_2) & \ldots & \psi_N(\boldsymbol{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_1(\boldsymbol{x}_N) & \psi_2(\boldsymbol{x}_N) & \ldots & \psi_N(\boldsymbol{x}_N) \end{vmatrix} \tag{11}$$

1.2 The Periodic Table

The fact that two or more fermions cannot occupy the same state gives rise to the periodic table of elements, since electrons are fermions are thus obey the Pauli exclusion principle. Electron orbital states are labeled by four quantum numbers $|n, \ell, m_\ell, m_s\rangle$, which come from solving the Schrodinger equation for a three-dimensional spherical potential.

- The **principle** quantum number n determines the *energy level* of the orbital, where higher n values correspond to greater average distances away from the nucleus. The allowed principle quantum numbers are only bounded from below by $n \geq 1$.
- The **Azimuthal** (or angular) quantum number ℓ determines the *shape* of the orbital:
  - $\ell = 0 \implies$ s orbital (spherical)
  - $\ell = 1 \implies$ p orbital (dumbbell)
  - $\ell = 2 \implies$ d orbital (cloverleaf)
  - $\ell = 3 \implies$ f orbital (more complex shape)

  and determines the angular momentum of the electron. Note that there are no theoretical restrictions on the existence of orbitals with azimuthal quantum number $\ell > 3$, but the current periodic table only extends to $\ell = 3$ at the heaviest element 118. The allowed values of ℓ are determined by the principal quantum number:

$$0 \leq \ell \leq n - 1 \tag{12}$$

- The **magnetic** quantum number m_ℓ determines the *orientation* of the orbital shape. The allowed values are between $-\ell$ and ℓ, including zero, meaning that for a given ℓ value, there are $2\ell + 1$ possible magnetic quantum numbers.

- The **spin** quantum number m_s determines whether the electron state is either spin up $|\uparrow\rangle$ or spin down $|\downarrow\rangle$. For a spin $\frac{1}{2}$ particle such as an electron, the only two allowed m_s values are $\pm\frac{1}{2}$.

Notice that the first three numbers $|n, \ell, m_\ell\rangle$ determine the nature of the orbital, while m_s is an intrinsic property of electrons. This is to say that there will be two allowed electron states for a single orbital state $|n, \ell, m_\ell\rangle$ due to the Pauli exclusion principle.

From these conditions on the four quantum numbers, we can construct the periodic table. Doing this for the first two principle quantum numbers:

$$n = 1, \ell = 0: \quad |1, 0, 0, +\frac{1}{2}\rangle \quad \& \quad |1, 0, 0, -\frac{1}{2}\rangle$$

These are the two allowed states in the $1s$ orbital (which is called $1s$ because $n = 1$ and $\ell = 0$). For the second principal quantum number:

$$\begin{aligned}
2s\ (n = 2, \ell = 0): \quad & |2, 0, 0, +\frac{1}{2}\rangle \quad \& \quad |2, 0, 0, -\frac{1}{2}\rangle \\
2p\ (n = 2, \ell = 1): \quad & |2, 1, -1, +\frac{1}{2}\rangle \quad \& \quad |2, 1, -1, -\frac{1}{2}\rangle \\
& |2, 1, 0, +\frac{1}{2}\rangle \quad \& \quad |2, 1, 0, -\frac{1}{2}\rangle \\
& |2, 1, 1, +\frac{1}{2}\rangle \quad \& \quad |2, 1, 1, -\frac{1}{2}\rangle
\end{aligned}$$

This tells us that there are 2 electrons allowed in the $2s$ orbital and 6 electrons allowed in the $2p$ orbital, making 8 total electrons allowed in the second energy level.

1.3 Condensation

1.4 Fermi Surface

Consider a free electron gas subject to the box potential

$$V(\boldsymbol{x}) = \begin{cases} 0, & 0 < x < \ell_x, 0 < y < \ell_y, 0 < z < \ell_z \\ \infty, & \text{else} \end{cases} \tag{13}$$

Solving the time-independent Schrodinger equation results in energy levels

$$E = \frac{\hbar^2 k^2}{2m} \tag{14}$$

where

$$k = \sqrt{k_x^2 + k_y^2 + k_z^2} \;\; \& \;\; k_i = \frac{\pi}{\ell_i}. \tag{15}$$

This result tells us that the allowed energy levels for this free electron gas are specified by the distance of the $\vec{k}$ vector away from the origin in k-space, which instead of spatial coordinates, has its points by the momenta of electrons (k_x, k_y, k_z). Since electrons obey the Pauli exclusion principle, we know that two electrons $|\uparrow\rangle, |\downarrow\rangle$ can occupy a single point in this space. This means that these electrons fill $k-$space as a sphere, since the wavevectors in Eq. (14) are isotropic, where the electrons with the lowest energy are found nearest to the center of the sphere. If the gas' particles were bosonic, they would instead pack at the lowest energy level and there would be no distinct shape in $k-$space.

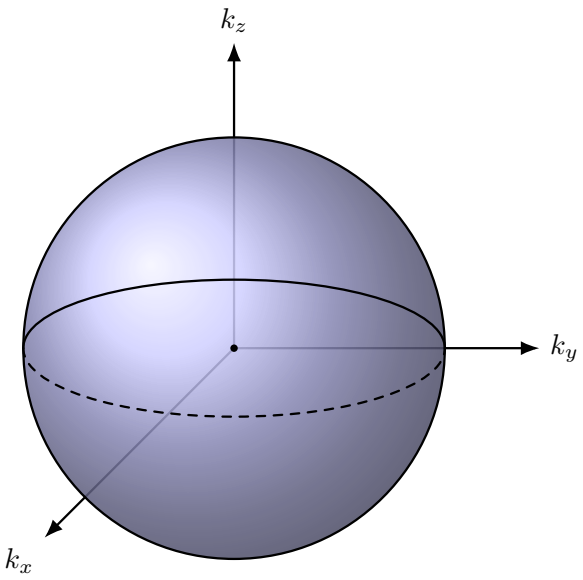


Figure 2: The simple Fermi surface of a free electron gas.

The sphere's size is determined by the magnitude of $\vec{k}$, which is governed by the electrons in the gas with the most energy. The boundary of this sphere

is called the *Fermi surface*, and electrons with energy lower than those found at the Fermi surface are defined to exist in the *Fermi sea*.

The Fermi surface is where the most potentially-mobile electrons in a gas or material are found. If energy is added to the system in the form of an external magnetic field, electric field, or heat, the highly-packed, low-energy electrons near the center of the Fermi sea will not be allowed to respond since every nearby state is already occupied. Only those electrons living near or on the Fermi surface will be allowed to respond to external excitations, since they are the only ones with empty states nearby.

The conductivity of metals is largely proportional to their Fermi surface area, since this is the quantity that dictates how many of the material's electrons are able to respond to external stimuli.

Conduction electrons are those that live at or near the Fermi surface.

This explanation of the Fermi surface comes from the simple example of a free electron gas, so it is reasonable to expect that as your potential becomes more complicated and realistic, the resulting energy levels will change, giving the Fermi surface a more complicated topology.

As a further example, consider a periodic potential $V(\boldsymbol{r}) = V(\boldsymbol{r} + \boldsymbol{R})$. Instead of the simple dispersion relation given in Eq. (14), this potential will result in very tightly-packed energy levels which essentially form continuous levels, called *energy bands*. The shapes of these independent bands will give rise to much more complicated Fermi surfaces.

2 Interacting Electrons and Nuclei

The starting point for solving for the dynamics and properties of a system of electrons and nuclei is the Hamiltonian:

$$\begin{aligned} H = &\frac{-\hbar^2}{2m_e}\sum_i^n \nabla_i^2 - \sum_I^N \frac{\hbar^2}{2M_I}\nabla_I^2 + \frac{1}{2}\sum_{i\neq j}^n \frac{e^2}{|\boldsymbol{r}_i - \boldsymbol{r}_j|} \\ &+ \frac{1}{2}\sum_{I\neq J}^N \frac{Z_I Z_J e^2}{|\boldsymbol{r}_I - \boldsymbol{r}_J|} - \sum_{i,I}^{n,N} \frac{Z_I e^2}{|\boldsymbol{r}_I - \boldsymbol{r}_i|}. \end{aligned} \tag{16}$$

The goal is to then solve the many-body Schrodinger equation for this Hamiltonian:

$$i\hbar\frac{\partial\Psi(\{r_i\},t)}{\partial t} = \hat{H}\Psi(\{r_i\},t). \tag{17}$$

We now define the density operator, which reduces solving the Schrodinger equation for each of the individual N particles to solving it for just a three-dimensional density. First defining a density:

$$\hat{n}(\boldsymbol{r}) = \sum_i^N \delta(\boldsymbol{r} - \boldsymbol{r}_i), \tag{18}$$

telling us that the density of a particle is sharply defined at a single point $\boldsymbol{r}_i$, we compute the number density of a many-body system by computing the expectation value of the number density operator:

$$\begin{aligned} n(\boldsymbol{r}) &= \langle\Psi|\hat{n}(\boldsymbol{r})|\Psi\rangle \\ &= \int \Psi^* d^3r_1 \dots d^3r_N \sum_i \delta(\boldsymbol{r} - \boldsymbol{r}_i) \int \Psi d^3r_1 \dots d^3r_N \\ &= \int \Psi^* d^3r_1 \dots d^3r_N \sum_i^N \int \Psi\delta(\boldsymbol{r} - \boldsymbol{r}_i) d^3r_1 \dots d^3r_N \\ &= N\int |\Psi|^2 d^3r_1 \dots d^3r_N \end{aligned} \tag{19}$$

3 Crystals

3.1 Lattice Structure

Crystals are solids with a periodic nature. Their periodic nature allows for a simple description of crystals, since the information necessary to describe it is only the set of repeating atoms, called a unit cell, and the translational symmetry of the unit cell, or how the unit cell spans the full material. The smallest possible unit cell, called the *primitive unit cell*, is described by primitive lattice vectors $\boldsymbol{a}_i$. These primitive lattice vectors span the full material under the translation vectors

$$\boxed{\boldsymbol{R}(n_1, n_2, n_3) = n_1\boldsymbol{a}_1 + n_2\boldsymbol{a}_2 + n_3\boldsymbol{a}_3} \tag{20}$$

where n_i are integers that describe the number of translations of the primitive unit vector $\boldsymbol{a}_i$ are needed to span the full material in the direction of $\boldsymbol{a}_i$.

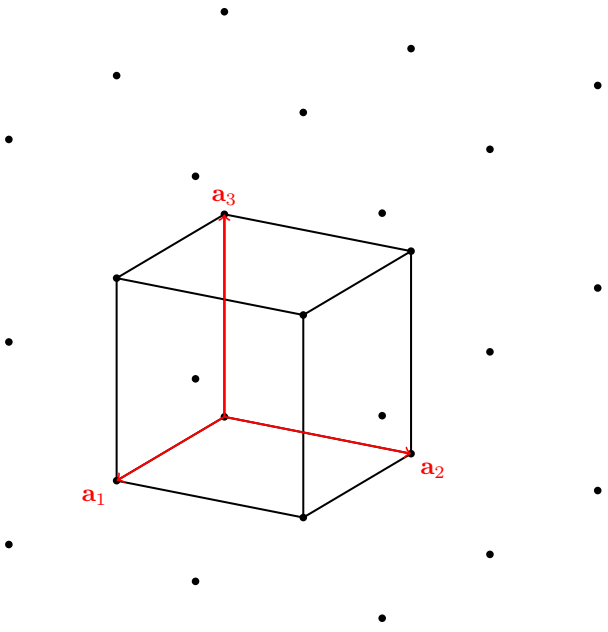


Figure 3: Primitive unit cell of a simple cubic lattice.

The number of translations in each direction N_i, or cells, needed to span the whole crystal determines its volume

$$\Omega_{\text{crystal}} = (N_1 \times N_2 \times N_3)\Omega_{\text{cell}}. \tag{21}$$

3.2 Wigner-Seitz Cells

The most symmetric primitive cell is called the Wigner-Seitz cell. It is constructed to contain only one lattice point inside of it and it defines the region of space closer to that central point than any other lattice point, which is very useful for simplifying electronic structure calculations. The $k-$space analog of the Wigner-Seitz cell is called the *first Brillouin zone*. Points with high degrees of symmetry in the Brillouin zone are given special names. The center point of the Brillouin zone, which is almost always the most symmetric, is labeled as Γ.

3.3 Bloch's Theorem

Before solving the Schrodinger equation for a periodic potential, we first define the translation operator

$$\begin{aligned} \hat{T}_{\boldsymbol{R}}\psi(\boldsymbol{r}) &= \psi(\boldsymbol{r}+\boldsymbol{R}) \\ &= \psi(\boldsymbol{r}+n_1\boldsymbol{a}_1+n_2\boldsymbol{a}_2+n_3\boldsymbol{a}_3) \\ &\equiv \psi(\boldsymbol{r}+\boldsymbol{A}\cdot\boldsymbol{n}) \end{aligned} \tag{22}$$

where $\boldsymbol{R}$ is the translation vector defined in Eq (20) and we have made the following vector simplifications:

$$\boldsymbol{A} = \begin{pmatrix} \boldsymbol{a}_1 \\ \boldsymbol{a}_2 \\ \boldsymbol{a}_3 \end{pmatrix} \quad \& \quad \boldsymbol{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix}. \tag{23}$$

The translation operator thus takes a state ψ at a position $\boldsymbol{r}$ and moves the state by some translation $\boldsymbol{R}$ in the crystal.

We can now see how the translation operator acts on the Hamiltonian of a periodic potential:

$$\begin{aligned} \hat{T}_{\boldsymbol{R}}\hat{H} &= \hat{T}_{\boldsymbol{R}}\left(\frac{\hat{p}^2}{2m}+\hat{V}(\boldsymbol{r})\right) \\ &= \frac{\hat{p}^2}{2m}+\hat{V}(\boldsymbol{r}+\boldsymbol{R}) \\ &\equiv \frac{\hat{p^2}}{2m}+\hat{V}(\boldsymbol{r}) = \hat{H}, \end{aligned} \tag{24}$$

where we used the fact that the potential of a periodic crystal is invariant under translations $\boldsymbol{R}$. We have thus shown that the translation operator

commutes with the Hamiltonian that features a periodic potential;

$$[\hat{T}_{\boldsymbol{R}}, \hat{H}] = 0, \tag{25}$$

so we know that the translation operator and Hamiltonian of a periodic potential have simultaneous eigenvectors.

We will first look for the eigenvalues of the translation operator:

$$\hat{T}_{\boldsymbol{R}}\psi(\boldsymbol{r}) = \lambda_{\boldsymbol{R}}\psi(\boldsymbol{r}). \tag{26}$$

Given that $\hat{T}_{\boldsymbol{R}}$ is an additive operator:

$$\hat{T}_{\boldsymbol{R}_1}\hat{T}_{\boldsymbol{R}_2}\psi(\boldsymbol{r}) = \psi(\boldsymbol{r} + \boldsymbol{R}_1 + \boldsymbol{R}_2) \equiv \hat{T}_{\boldsymbol{R}_1+\boldsymbol{R}_2}, \tag{27}$$

this tells us that its eigenvalues must be additive as well:

$$\begin{aligned} &\lambda_{\boldsymbol{R}_1}\lambda_{\boldsymbol{R}_2}\psi(\boldsymbol{r}) = \lambda_{\boldsymbol{R}_1+\boldsymbol{R}_2}\psi(\boldsymbol{r}) \\ &\implies \lambda_{\boldsymbol{R}_1}\lambda_{\boldsymbol{R}_2} = \lambda_{\boldsymbol{R}_1+\boldsymbol{R}_2} \end{aligned} \tag{28}$$

A function that satisfies this property is the exponential function, so we first express the eigenvectors of the translation operator as:

$$\lambda_{\boldsymbol{R}} = e^{s\boldsymbol{A}\cdot\boldsymbol{n}} \tag{29}$$

where $s \in \mathbb{C}$.

Now using the normalization of the wavefunction within a volume Ω_{cell} determine s:

$$\begin{aligned} 1 &= \int_\Omega |\psi(\boldsymbol{r})|^2 d^3r \equiv \int_\Omega |\hat{T}_{\boldsymbol{R}}\psi(\boldsymbol{r})|^2 d^3r \\ &= |\lambda_{\boldsymbol{R}}|^2 \int_\Omega |\psi(\boldsymbol{r})|^2 d^3r \\ &\implies |\lambda_{\boldsymbol{R}}|^2 = |e^{s\boldsymbol{A}\cdot\boldsymbol{n}}|^2 = 1 \implies s = ik \end{aligned} \tag{30}$$

where k is some real number to be determined.

Using Eq. (26), we have shown that

$$\hat{T}_{\boldsymbol{R}}\psi(\boldsymbol{r}) = e^{ik\boldsymbol{A}\cdot\boldsymbol{n}}\psi(\boldsymbol{r}) \tag{31}$$

Since the translation operator and the periodic Hamiltonian share eigenvectors, we have demonstrated that the shared eigenvectors should include the plane waves from the translation operator we have just derived and the

eigenvalues of the Hamiltonian itself, which we are commonly denoted as $u(\boldsymbol{r})$. Then for solving the Schrodinger equation

$$\hat{H}\psi_n(\boldsymbol{r}) = E_n\psi_n(\boldsymbol{r}), \tag{32}$$

we have that

$$\psi_n(\boldsymbol{r}) = e^{ik\boldsymbol{A}\cdot\boldsymbol{n}}u_n(\boldsymbol{r}), \tag{33}$$

which is called **Bloch's theorem**. The $u(\boldsymbol{r})$ functions, called Bloch functions, have the same periodicity as the lattice.

Bloch's theorem is very important in determining the energy level structure of a crystalline material, since it tells us that the whole material shares the same spectrum as one of the cells.

3.4 Reciprocal Space

Since a quantity $f(\boldsymbol{r})$, such as number density, is periodic in a crystalline material:

$$\hat{T}_{\boldsymbol{R}}f(\boldsymbol{r}) = n(\boldsymbol{r}+\boldsymbol{R}) = f(\boldsymbol{r}), \tag{34}$$

it can be represented by Fourier transforms in terms of Fourier components at wavevectors $\boldsymbol{k}$. Assuming a finitude of the crystal volume, this imposes the usual boundary conditions that

$$\boldsymbol{k}\cdot\boldsymbol{a}_i = \frac{2\pi n}{N_i} \tag{35}$$

where $n \in \mathbb{N}$. This allows us to represent one of these periodic quantities as

$$f(\boldsymbol{k}) = \frac{1}{\Omega}\int_\Omega f(\boldsymbol{r})e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}d^3r \tag{36}$$

Applying the translation operator:

$$\begin{aligned}
\hat{T}_{\boldsymbol{K}}f(\boldsymbol{k}) &= \frac{1}{\Omega}\int_\Omega \hat{T}_{\boldsymbol{R}}\left(f(\boldsymbol{r})e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\right)d^3r \\
&= \frac{1}{\Omega}\int_\Omega f(\boldsymbol{r})\hat{T}_{\boldsymbol{R}}e^{i\boldsymbol{k}\cdot\boldsymbol{r}}d^3r \\
&= \frac{1}{\Omega}\int_\Omega f(\boldsymbol{r})e^{i\boldsymbol{k}\cdot(\boldsymbol{r}+\boldsymbol{R})}d^3r \\
&= \sum_{n_1,n_2,n_3} e^{i\boldsymbol{k}\cdot\boldsymbol{R}(n_1,n_2,n_3)}\frac{1}{\Omega}\int f(\boldsymbol{r})e^{i\boldsymbol{k}\cdot\boldsymbol{r}}d^3r.
\end{aligned} \tag{37}$$

Just as the crystal lattice positions could be represented as a repeating set of basis vectors $\{\boldsymbol{a}_i\}$, which would span the full material under the translation vector defined in Eq. (20), the reciprocal space, or $k-$space, can be represented in a basis $\{\boldsymbol{b}_i\}$ that spans the full space under the translation

$$\boldsymbol{G}(m_1, m_2, m_3) = m_1\boldsymbol{b}_1 + m_2\boldsymbol{b}_2 + m_3\boldsymbol{b}_3, \tag{38}$$

where

$$\boldsymbol{a}_i \cdot \boldsymbol{b}_j = 2\pi\delta_{ij}. \tag{39}$$

The last relation guarantees that reciprocal space features the same periodicity as real space.

3.5 Wyckoff Sites

Definition 1. A **lattice** Λ is a set of all sums of linearly-independent primitive lattice (basis) vectors $\boldsymbol{a}_i$ such that

$$\Lambda = \left\{ \sum_{i=1}^{3} m_i | m_{\in} \mathbb{Z}\boldsymbol{a}_i \right\} \tag{40}$$

where $m_i \in \mathbb{Z}$ tell you how many repetitions of the basis vectors are needed to span the full lattice.

Definition 2. A **space group** $\mathcal{G}$ describes the rigid transformations that a lattice can undergo without changing the periodic structure.

A space group is therefore all operations of the primitive cell that preserve the isometry of the primitive lattice vectors.

Definition 3. The **orbit** $\mathcal{O}$ of an atom is the set of all infinitely many positions an atom can have under the space group symmetry operations (the operations allowed by the unit cell's symmetry operations that preserve the basis vector lengths). Formally, for a point $X \in \mathbb{E}^3$, the **orbit** of X under $\mathcal{G}$ is

$$\mathcal{O} = \mathcal{G}(X) = \{g(X) | g \in \mathcal{G}\} \tag{41}$$

Definition 4. The **site symmetry groups** $\mathcal{S} \subseteq \mathcal{G}$ are the symmetry operations that leave X unchanged.

Formally, if $g(X) = X$ for $g \in \mathcal{G}$, then $g \in \mathcal{S}_X \subseteq \mathcal{G}$.

Definition 5. Two points X and Y belong to the same **Wyckoff site** with respect to $\mathcal{G}$ if their respective point symmetry groups $\mathcal{S}_X$ and $\mathcal{S}_Y$ are conjugate subgroups of of $\mathcal{G}$; that is if there exists a $g \in \mathcal{G}$ such that

$$\mathcal{S}_Y = g^{-1}\mathcal{S}_X g \text{ for } g \in \mathcal{G}, \tag{42}$$

meaning if you can form $\mathcal{S}_Y$ from $\mathcal{S}_X$ by applying another space group operation $g \in \mathcal{G}$.

4 Band Theory

4.1 Band Structure Basics

We first solve the electron Schrodinger equation for a periodic potential:

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\boldsymbol{r}) + V(\boldsymbol{r})\psi(\boldsymbol{r}) = E\psi(\boldsymbol{r}). \tag{43}$$

We know from Bloch's theorem that wavefunctions under periodic potentials can be written as

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{R}}u_{n\boldsymbol{k}}(\boldsymbol{r}) \tag{44}$$

where $e^{i\boldsymbol{k}\cdot\boldsymbol{R}}$ is the plane wave envelope and $u_{n\boldsymbol{k}}(\boldsymbol{r})$ is a basis wavefunction that has the same periodicity as the lattice. Plugging this wavefunction in, derivatives act as

$$\begin{aligned}\nabla\left(e^{i\boldsymbol{k}\cdot\boldsymbol{R}}u_{n\boldsymbol{k}}(\boldsymbol{r})\right) &= u_{n\boldsymbol{k}}(\boldsymbol{r})e^{i\boldsymbol{k}\cdot\boldsymbol{R}}i\boldsymbol{k} + e^{i\boldsymbol{k}\cdot\boldsymbol{R}}\nabla u_{n\boldsymbol{k}}(\boldsymbol{r}) \\ &\implies \nabla \to \nabla + i\boldsymbol{k}\end{aligned} \tag{45}$$

$$\implies -\frac{\hbar^2}{2m}(\nabla + i\boldsymbol{k})^2 u_{n\boldsymbol{k}}(\boldsymbol{r}) + V(\boldsymbol{r})u_{n\boldsymbol{k}}(\boldsymbol{r}) = Eu_{n\boldsymbol{k}}(\boldsymbol{r}). \tag{46}$$

We now have an eigenvalue problem for each $\boldsymbol{k}$ on a unit cell with periodic boundary conditions. In other words, for each $\boldsymbol{k}$, there could be various energy levels $E_1(\boldsymbol{k}), E_2(\boldsymbol{k}), \ldots, E_n(\boldsymbol{k})$, which could give rise to *energy bands* if these levels have uneven spacing.

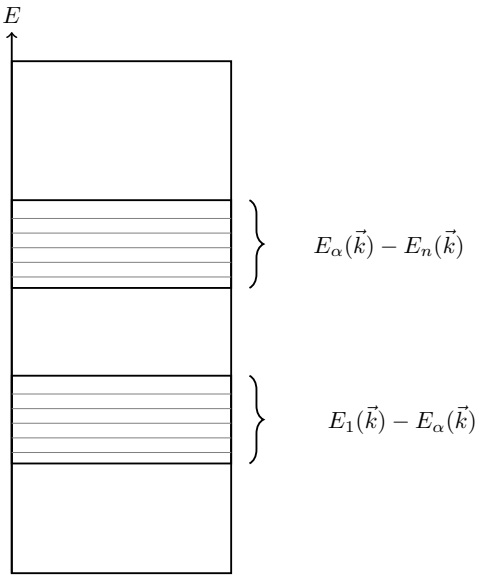


Figure 4: Two energy bands featuring gaps above and below.

4.2 Kronig Penney Model

The following is a canonical example of a simple 1D potential that gives rise to exactly solvable energy bands. Consider a periodic potential of Dirac delta functions

$$V(x) = \alpha \sum_{j}^{N-1} \delta(x - ja), \tag{47}$$

shown in Fig. 5.

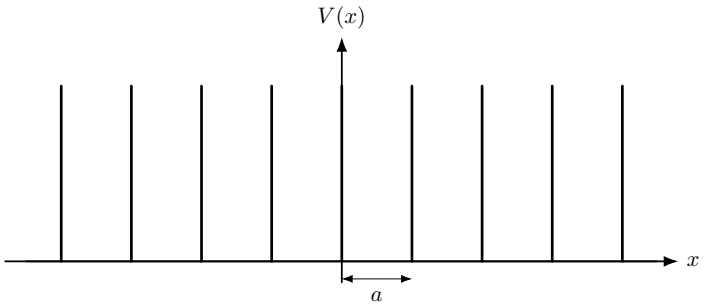


Figure 5: The "Dirac comb" potential of the Kronig Penney model.

By Bloch's theorem, we only need to solve the Schrodinger equation inside one of the potential's cells, so we choose the region $0 < x < a$. Inside this cell, solutions look like:

$$\psi(x) = A\sin(kx) + B\cos(kx). \tag{48}$$

We then use Bloch's theorem in 1D,

$$\psi(x + R) = e^{iRa}\psi(x), \tag{49}$$

to construct the wavefunction for any x. Considering the cell to the left:

$$\psi(x) = e^{-iRa}\Big[A\sin(k(x + a)) + B\cos(k(x + a))\Big] \qquad (-a < x < 0). \tag{50}$$

Using boundary conditions to solve for A and B results in the equation

$$\cos(Ra) = \cos(ka) + \frac{m\alpha}{\hbar^2 k}\sin(ka). \tag{51}$$

We then define $z = ka$ and $\beta = \frac{m\alpha a}{\hbar^2}$. Then

$$\cos(Ra) = \cos(z) + \beta\frac{\sin(z)}{z}. \tag{52}$$

The right-hand side of this equality must be bounded between $[-1, 1]$ to produce valid solutions since $\cos(Ra) \in [-1, 1]$. How the valid solutions to this expression give rise to the idea of energy bands is shown in Fig. 6

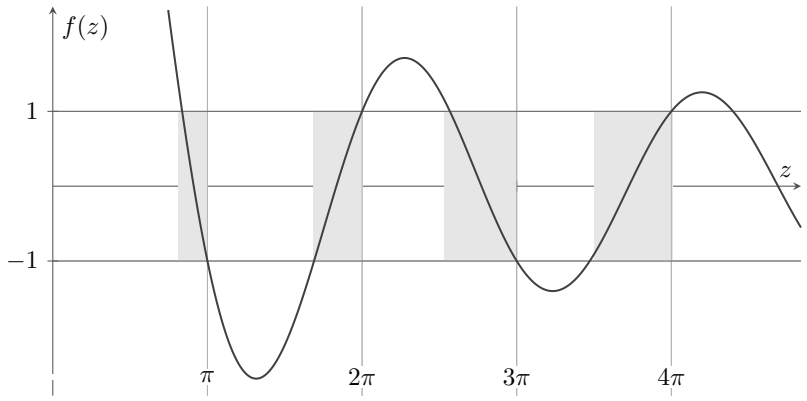


Figure 6: Where $f(z) \in [-1, 1]$ produces valid solutions to Eq. (52) is shown in the shaded regions. Inside these regions exist valid energy levels for the wavefunction. The grey shaded regions are thus called *energy bands* and the spaces between them are called *band gaps*.

These bands give rise to the definitions of materials:

1. If the top-most band, called the *valence band*, is only partially filled, it takes little energy to excite an electron to a higher level, called the *conduction band*. These materials are called **conductors**.

2. If the valence band is completely filled, it takes much more energy to excite an electron to the conduction band, since it must cross a band gap. These materials are called **insulators**.

3. If the band gaps are narrow enough, however, thermal energy will be enough to excite electrons through a band gap from a valence to conduction band. These materials are called **semiconductors**.

4.3 Topological Materials

A topological material is one whose band structure can not be smoothly deformed into an atomic insulator without closing the energy gap, defined by the Fermi energy. In other words, a material is topological if its band structure belongs to a non-trival topological class, defined by its Chern or $\mathbb{Z}_2$ number (see later section).

A very important topological class are *topological insulators*. These are materials whose bulk band structure is gapped (it is an electronic insulator), but whose boundary features conducting states that are guaranteed ("topologically protected") by the non-trivial band structure there. Topologically protected here means that the conduction at the boundary of these materials cannot be destroyed by added defects or impurities in the material.

4.4 Berry Curvature

4.5 Topological Invariants

Discuss Chern number and $\mathbb{Z}_2$.

The Chern number is a topological invariant used to classify which topological class a material belongs to (specifically, it classifies chiral edge states). It is defined as the total Berry curvature flux through a closed, two-dimensional manifold (such as the Brillouin Zone):

$$C = \frac{1}{2\pi} \int_S \boldsymbol{B}(\boldsymbol{R}) dS \in \mathbb{Z}. \tag{53}$$

Chrial (one-way) edge states are electron transport channels at the boundary of a material (allowing the material to conduct at the boundary). The Chern number of a vacuum is always 0, so if a material with a non-zero Chern number n is placed inside a vacuum, the Chern number must transition from n to 0 over the boundary of the material. There must therefore be a region on the boundary of the material where the energy gap is closed, making the boundary of the material conducting.

The canonical topological insulator is bismuth selenide Bi_2Se_3.

Note that $\mathbb{Z}_2$ number of a material is computed from the irreducible representation (explain this)

4.6 Quantum Hall Effect and Edge States

5 Lattice Vibrations

5.1 Phonons

5.2 Dispersion

5.3 Electron-Phonon Coupling

5.4 Energy above Hull

6 Computation and Data

6.1 Crystal Structure Databases

- **Inorganic Crystal Structure Database** (ICSD):
  - Founded in 1978 in Germany to compile all of the existing crystal structure data, stretching all the way back to 1913.
  - Crystal structure data comes from published data on measured x-ray crystallography or neutron diffraction data.
  - As of August 2025, there are 318,000 entries
  - Typically used when experimental validity of the materials is important
- **Materials Project** (MP):
  - Created in 2011 by Kristin Persson (postdoc) and Gerbrand Ceder at MIT in 2011.
  - Material properties in this database are computed by DFT, not experimentally measured.
  - Great for automated, high-throughout, large-scale workflows since the recorded data is very standardized.
- **Open Quantum Materials Database**(OQMD)
  - Developed in 2010 by Chris Wolverton's group at Northwestern University.
  - Focuses on thermodynamic stability of materials, and thus is used if one wants to determine whether a proposed compound will be stable when compared against decomposition into competing phases.
- **Topological Materials Database**:
  - Came out of Bernevig's group's 2019 paper that classified whether a material is topological or not based on its band structure.
  - Features the band structure of known topological materials, topological classifiers, and symmetry indicators.

– Also features a machine learning classifier tool for whether a proposed material will be topological or not.

7 Experimental Techniques

7.1 Basic Definitions

- "Purity" of a sample: how close the experimental realization of a sample is to the predicted compositions and ratios.
- "Stoichiometric material" is one whose theoretical composition is exactly observed when synthesized (no substitutional disorder).
- Annealed sample: a material that has undergone significant heating and then cooling to reduce disorder.

7.2 X-Ray Crystallography

8 Superconductivity

8.1 Experimental Fast Facts

- Highest critical temperatures:
  - At **ambient pressure**: mercury-based copper-oxide ceramic (Hg – 1223) at $T_c = 151K$
  - Under **extreme pressure** (150+ GPa): lanthanum superhydride (LaH_{10}) at $T_c = 250K$
- **First superconductor**:
  - Discovered in April 8th, 1911 by Dutch physicist Heike Kamerlingh Onnes (famous low-temperature physicist who first produced liquid helium)
  - Discovered in mercury (Hg) at $T_C = 4.2K$

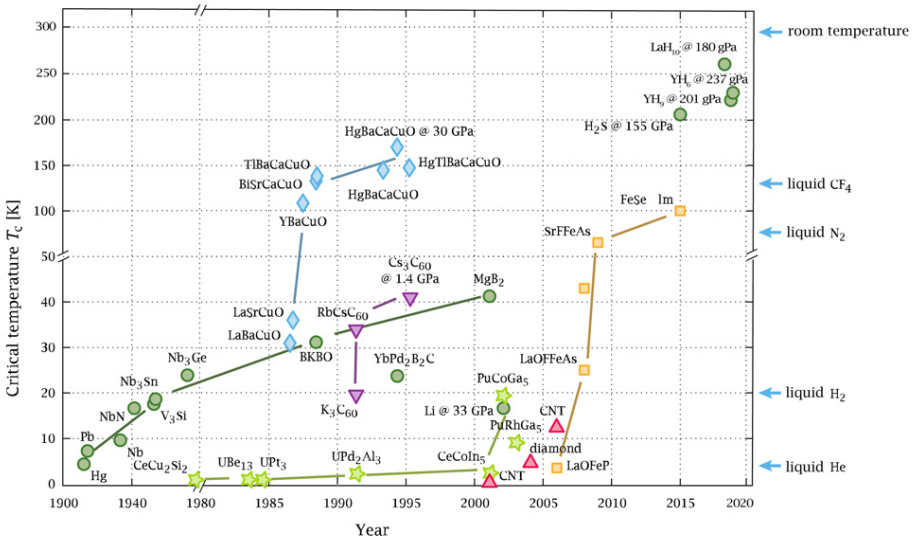


Figure 7: A timeline of the critical temperatures for experimentally-validated superconductors. Dark green circles represent BCS materials, light green stars represent heavy fermions, light blue diamonds are cuprates. Taken from superconductivity Wikipedia page.

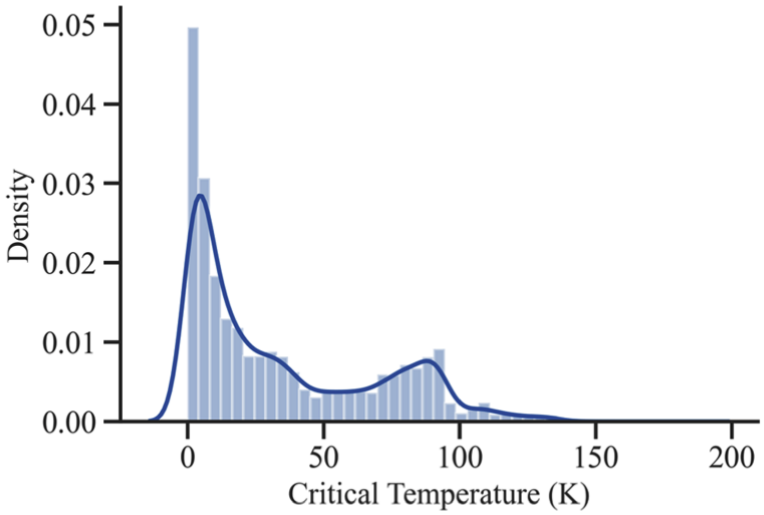


Figure 8: The probability density over T_c of known superconducting materials from the SUPERCON database as of 2023. Grabbed from [1].

8.2 Practical Applications

Explain how MRI machines use them.

References

[1] Jiahao Yu et al. "Prediction of the Critical Temperature of Superconductors Based on Two-Layer Feature Selection and the Optuna-Stacking Ensemble Learning Model". In: *ACS Omega* 8.3 (2023), pp. 3078–3090. DOI: 10.1021/acsomega.2c06324. URL: https://doi.org/10.1021/acsomega.2c06324.